Elias Bitsch

Mixed Reality for Human-Robot Collaboration | XR Training Systems | User Studies with Industry Partners

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Research Profile

Interactive systems for human-robot collaboration, built end to end and then tested in controlled user studies with industry participants. Mixed reality is the centre of the work, bare-hand authoring and teleoperation across Unity and ROS 2, but the same question is pursued through every channel an operator actually uses: touch interfaces for supervising mobile robots, tangible instruments, and haptic machine controls. Both the teleoperation stack and the measurement layer around it are built and running, and can be pointed at a host lab's own research question rather than rebuilt.

Education

UAS Technikum Wien — Vienna, Austria
M.Sc. in Robotics Engineering — Sep 2025 – expected 2027

- Specialisation: Robot Learning
- Master's thesis in preparation on mixed-reality training and skill acquisition in human-robot collaboration

UAS Technikum Wien — Vienna, Austria
B.Sc. in Mechatronics and Robotics — 2023 – 2025

- Robotics software in C++, Python, ROS/ROS 2, Gazebo, RViz and OpenCV, deployed in Docker
- Hardware development: PCB design (EasyEDA), CAD and 3D-printed prototypes (SolidWorks, Fusion 360)

HTL Eisenstadt, Higher Technical College — Eisenstadt, Austria
Five-year secondary programme in Mechatronics, GPA 1.86 (1 = best) — 2016 – 2021

Publications and Presentations

Bitsch, E., Ovdiienko, V., Stix, P., Saliger, A., Angel, N., Orsolits, H. (2026). *MetaMove: Bare-Hand Mixed-Reality Teleoperation of a Collaborative Robot with a Distance-Based Speed-Scaling Safety Layer.* XR-SPro Workshop, IEEE ISMAR 2026 Adjunct Proceedings (ISMAR-Adjunct), Bari, Italy. **Accepted.**

Demonstration. *Human–Robot Interaction for Casting Parts and Defect Annotation*, Applied AI Conference (AAIC) 2026, shown in a joint AIT and Fraunhofer session. Built and presented by the author; no accompanying paper.

Research Experience

AIT Austrian Institute of Technology — Vienna, Austria
Technical Assistant, Center for Technology Experience — Aug 2025 – present
Research and prototyping in human-robot interaction

- **Mixed reality (Unity, C#, Meta Quest):** hand-tracked interaction, passthrough, spatial anchors and digital twins, built and deployed as standalone Quest applications
- **Robotics middleware (ROS 2):** MoveIt and MoveIt Servo, Nav2 and behaviour trees, real-time bridges to industrial robot controllers, containerised with Docker
- **Web-based operator interfaces (React, Three.js, Node.js):** 3D supervision dashboards over a Node bridge to ROS 2, usable from a browser without robotics knowledge
- **Embedded feedback hardware (ESP32, IMU, haptics, LEDs):** handheld and wearable instruments from PCB and firmware through to the interaction design

- **Human-robot interaction research:** within-subject designs with controlled failure injection, multimodal feedback across visual, auditory and haptic channels, observational analysis and questionnaires with industry participants
- **Note:** the individual projects are covered by client and institute confidentiality; details on request

UAS Technikum Wien, MetaMove project — Vienna, Austria
Semester project, first author of the resulting publication — 2026

- **Bare-hand XR teleoperation:** of an ABB GoFa collaborative robot, Unity to ros-tcp-connector to ROS 2 with MoveIt Servo, an EGM bridge and OmniCore at 250 Hz
- **Safety layer:** distance-based speed scaling that reduces robot velocity as the operator approaches, plus a hardware-free RobotStudio simulation path
- **Outcome:** accepted as a short paper at the XR-SPro workshop, IEEE ISMAR 2026; because the simulation path needs no robot, the stack can be stood up in a new lab before hardware time is available

Professional Experience

Elektrobit Austria GmbH — Vienna, Austria
Software Developer (internship) — Feb – Jun 2025

- **Automotive middleware:** designed and implemented an extension of the SOME/IP protocol that carries and honours quality-of-service parameters in distributed AUTOSAR Classic systems, so that services are discovered and prioritised across ECUs
- **Outcome:** deployed on automotive test hardware and tested successfully

Mars Incorporated — Austria
Service and maintenance technician (internship) — Aug 2018, Aug 2019

- **Robot maintenance:** SCARA robots and welding of robot components, including an articulated-arm gripper

Teaching, Leadership and Service

IEEE ICRA 2026 — Vienna, Austria
Student volunteer — 2026

UAS Technikum Wien — Vienna, Austria
Team lead, Sumo-Bot competition — 2023 – 2025

- **Team lead:** guided a multidisciplinary team of seven through design, construction and programming of a competition robot; mentored members in programming, hardware design and troubleshooting

UAS Technikum Wien — Vienna, Austria
Technical guide, open day — 2023 – 2025

- **Outreach:** guided prospective students through campus and laboratories and presented the degree programmes

Technical Skills

Robotics ROS/ROS 2, Gazebo, RViz, MoveIt, Nav2, behaviour trees, OPC UA, ABB RAPID/EGM, pymycobot
XR Unity, Meta XR SDK, OpenXR, hand tracking, passthrough, spatial anchors, digital twins
Languages C, C++, Python, Bash, JavaScript, TypeScript, SQL
Web and infrastructure React, Three.js, Next.js, Node.js, Docker, Git, CI/CD, AWS
Hardware ESP32/PlatformIO, Arduino, PCB design (EasyEDA), SolidWorks, Fusion 360, 3D printing, soldering
Vision and ML OpenCV, vision-language models, DeepFace, Whisper

Certificates

Dassault Systèmes, Mechanical Design Certificate, UAS Technikum Wien, 2023
Summer School “Quantum Technologies”, UAS Technikum Wien, 2023

Languages German (native) · English (fluent)